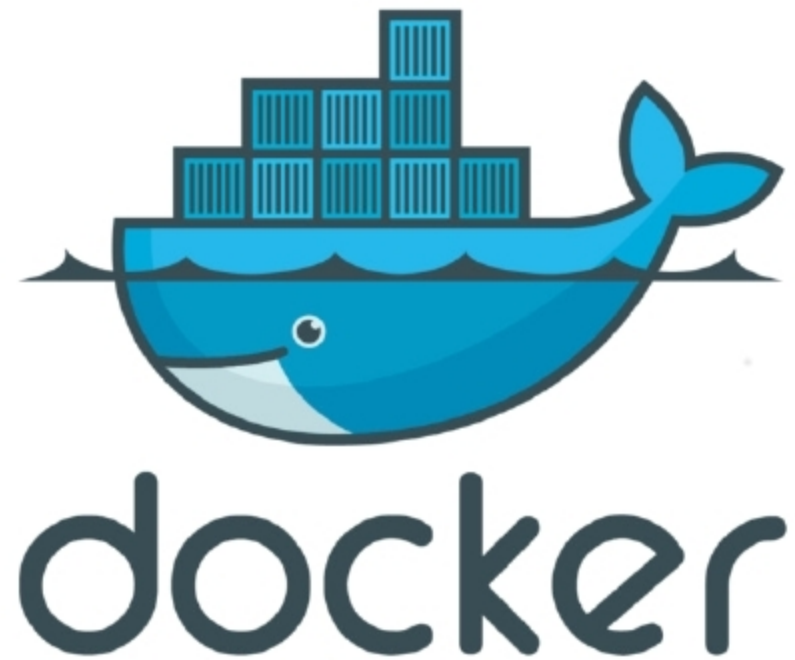


Configuring Jenkins on Docker

This article gives an overview of both Jenkins and Docker. It also includes a tutorial on how to configure Jenkins on Docker.



Jenkins is an open source, self-reliant automation server initially developed for the implementation of continuous integration (CI), a development practice. From Jenkins 2 onwards, there has been an exceptionally abundant plugin ecosystem to integrate with the toolchain already in use in an organisation. This enables Jenkins to not only go beyond CI but also implement various other practices such as continuous code inspection, continuous delivery, continuous testing, continuous deployment, and so on. Practically, this tool can automate almost all the steps followed in the practices mentioned and their tools, in the desired order. It wraps up all the tasks in a particular sequence and divides them into various DevOps stages known as the pipeline. Thus, it promotes end-to-end automation of the IT life cycle.

Jenkins is a Web server application that can be deployed as a servlet in a Java servlet application/container server such as GlassFish, Wildfly, IBM WebSphere, and so on. Jenkins also

provides a built-in Jetty server, a Java servlet application server, to run as a standalone server. In order to have a Jenkins setup running on a system, the only thing required is the Java 8 Runtime Environments (JRE 8). The minimum hardware requirements are 256MB of RAM and 1GB disk space, whereas the recommended requirements are 1GB or more of RAM and 10GB or more (as per the scale of the project) of disk space.

The language in which Jenkins is written is Java, which makes it easier to design for installing/running it on various operating systems such as MacOS, Windows, Linux and their different flavours such as Ubuntu, Red Hat, Debian as well as platforms like Docker, Microsoft Azure, AWS EC2, etc.

Jenkins' features

Jenkins started out as a simple CI server

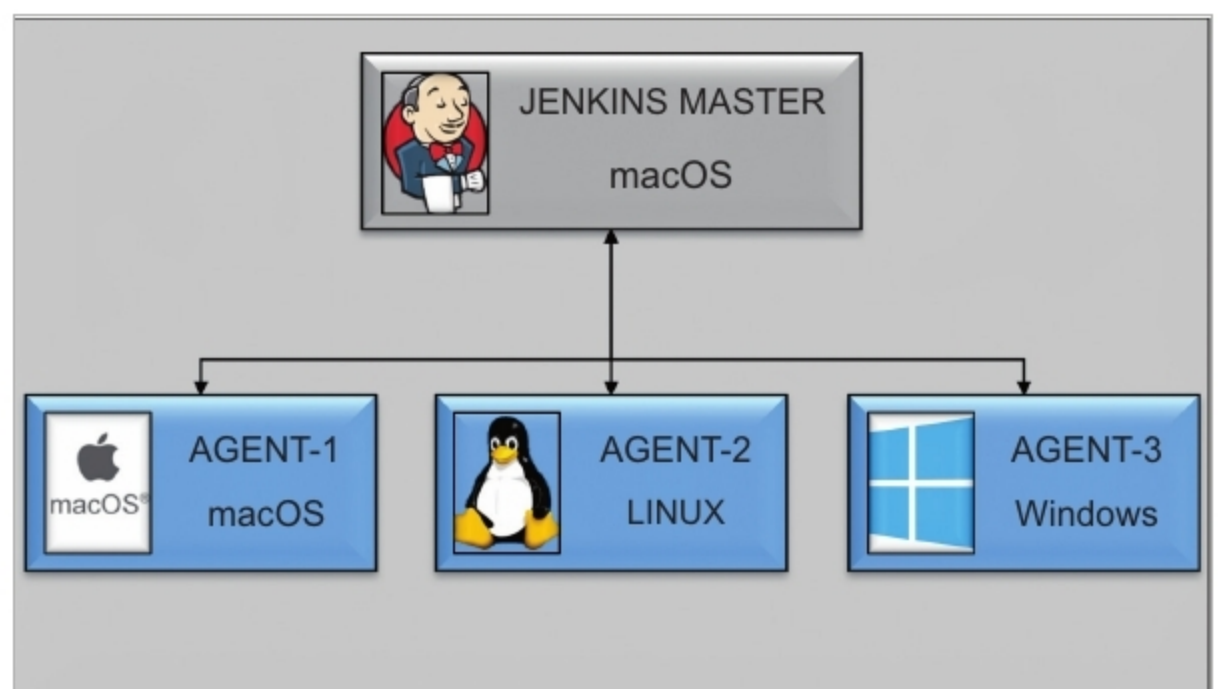


Figure 1: Jenkins master-agent architecture